Karan S Suvarna

Mangalore | suvarnakaran77@gmail.com | 8073788715 |
LinkedIn | GitHub

Objective

MCA graduate with practical experience in software development, backend development, and AI-powered applications. Passionate about designing scalable, efficient software solutions and committed to contributing to innovative projects while continuously enhancing technical skills and staying current with emerging technologies.

Experience

Software Developer Intern – Softionik Solutions (OPC) Pvt. Ltd. **Feb 2026 – May 2026**

- Worked on full-stack MERN application development following industry-level SDLC practices.
- Gained hands-on experience in frontend, backend, API design, and database integration.
- Collaborated on real-world problem solving, testing, and deployment workflows.

Projects

DecorMatch AI – Intelligent Interior Styling Assistant

Developed an AI-powered interior design recommendation system that analyzes room images and generates personalized furniture and décor suggestions. Implemented EfficientNetV2 for interior style classification (89.95% accuracy), YOLOv8 for object detection, and K-Means clustering for dominant color extraction. Built a scalable Flutter, Node.js, and FastAPI-based architecture and integrated ARCore-powered 3D visualization to preview recommended furniture in real-world spaces.
Tech Stack :Flutter, Dart, Python, FastAPI, Node.js, TensorFlow, YOLOv8, EfficientNetV2, OpenCV, Scikit-learn, Firebase, ARCore.

Vistone AI - Powered Colour Palette Matcher

Built an AI-driven skin analysis system that determines a user's Monk Skin Tone (1–10) and undertone (Warm, Cool, Neutral) using advanced computer vision techniques. Implemented precise facial-region masking with MediaPipe Face Mesh, illumination-correction pipelines, LAB colour analysis, and Gaussian Mixture Models for highly accurate dominant skin-colour extraction. Integrated a JSON-based colour recommendation engine to generate top 3 suitable colours and top 3 colours to avoid for personalized fashion and styling use cases.
Tech Stack : Python, OpenCV, MediaPipe, scikit-learn (GMM), Color Science (LAB/ΔE), Image Processing, JSON-based Colour Mapping.

iCar – Smart Carpooling Web Application (Internship Project)

Developed a full-stack MERN application for ride sharing, enabling users to offer, search, and book carpool rides. Built responsive UI using React.js and designed RESTful APIs with Node.js and Express.js for authentication, ride management, and booking. Integrated MongoDB for data handling and implemented JWT-based authentication. Added features like smart ride matching, in-app chat, and a dashboard to track cost savings and environmental impact.
Tech Stack : MongoDB, Express.js, React.js, Node.js, JWT.

Technical Skills

Languages: Python, Java, JavaScript, C, SQL.

Frameworks & Tools: Flask, Node.js, React.js, Flutter, Spring Boot, HTML, CSS.

Databases: MongoDB, MySQL, Firebase.

Education

N.M.A.M. Institute of Technology, Nitte . 2024 – 2026
Master of Computer Applications

Canara Degree College . 2021 – 2024
Bachelor of Computer Applications